Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Fiscal Policy — qualifier level (Listas 5–6)

Solutions and grading rubric

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — General equilibrium, hours and taxes

(Kurlat ch. 9; Romer ch. 2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) *General equilibrium × labor supply.* A two-period economy has a fixed capital stock K and no investment. Households have $u(c) + \psi \ln l$ each period, $u = c^{1-\sigma}/(1-\sigma)$, one unit of time; firms produce $Y = AK^{\alpha}L^{1-\alpha}$ competitively. In equilibrium, with $L = 1/3$, $\sigma = 2$ and $\alpha = 1/3$, the elasticity $d\ln L/d\ln A$ is:

A. -0.46

B. 0

C. $+0.46$

D. -1.00

Answer: A

Step — equilibrium conditions. Goods clearing without investment: $c = Y$. Firm: $w = (1-\alpha)Y/L$. Household: $\psi/(1-L) = w\,u'(c)$. Combining: $\dfrac{\psi}{1-L} = \dfrac{(1-\alpha)Y^{1-\sigma}}{L}$.

Step — log-differentiation. $\frac{L}{1-L}d\ln L = (1-\sigma)d\ln Y - d\ln L$ and $d\ln Y = d\ln A + (1-\alpha)d\ln L$: $\frac{d\ln L}{d\ln A} = \frac{1-\sigma}{L/(1-L)+\alpha+(1-\alpha)\sigma} = \frac{-1}{0.5+1/3+4/3} = -0.46$. With $\sigma > 1$ the income effect dominates; B is the log case.

Ref: Kurlat (2020), §9.1, pp. 165–168; Lista 5, Q1(e).

b) (3 points) *General equilibrium × growth.* In the continuous-time Ramsey model with $f(\tilde{k}) = \tilde{k}^{1/3}$, $\rho = 0.03$, $\sigma = 2$, $g = 0.02$, $n = 0.01$, $\delta = 0.05$, the steady-state gross saving rate is:

A. 33.3%

B. 22.2%

C. 26.7%

D. 16.7%

Answer: B

Step — modified golden rule. $\dot{\hat{c}} = 0$: $f'(\tilde{k}^*) = \rho + \delta + \sigma g = 0.12$; $s = \frac{(n+g+\delta)\tilde{k}}{f(\tilde{k})} = \frac{(n+g+\delta)\alpha}{f'(\tilde{k})} = \frac{0.08/3}{0.12} = 22.2\%$, below the Golden Rule's $\alpha = 33.3\%$ (A). C uses $\sigma = 1$.

Ref: Romer (2012), §2.5 and problem 2.6; Kurlat (2020), §9.3, p. 177.

c) (2 points, T or F) *General equilibrium × fiscal policy.* In the Ramsey model, a permanent, unanticipated rise in government purchases financed by lump-sum taxes lowers consumption one for one on impact and leaves capital unchanged forever.

Answer: T

Step — why. $\dot{\tilde{k}} = f(\tilde{k}) - \hat{c} - \hat{G} - (n+g+\delta)\tilde{k}$ and the Euler equation does not involve $\hat{G}$: the new steady state has the same $\tilde{k}^*$ and $\hat{c}^*$ lower by $\Delta\hat{G}$; the saddle path shifts down by $\Delta\hat{G}$, so $\hat{c}$ jumps by $-\Delta\hat{G}$ and there is no transition.

Ref: Romer (2012), §2.7, pp. 71–76.

d) (2 points, T or F) *General equilibrium × taxes.* A permanent tax on capital income whose revenue is rebated lump-sum leaves the steady-state capital stock unchanged, because the rebate gives the revenue back.

Answer: F

Step — wedge. The rebate removes the income effect, not the wedge: $(1-\tau)(f'-\delta) = \rho + \sigma g$, so $f'(\tilde{k}^*)$ rises and $\tilde{k}^*$ falls.

Ref: Romer (2012), problem 2.10; Kurlat (2020), Exercise 9.6.

Question 2. (15 points) Discursive. *General equilibrium × labor supply × taxes.* An infinite-horizon economy without growth: households maximize $\int_0^\infty e^{-\rho t}[\ln c + \psi \ln(1-h)]\,dt$, own the capital and rent it to firms; firms produce $Y = K^\alpha h^{1-\alpha}$; capital depreciates at δ; labor income is taxed at τ and the revenue is rebated lump-sum. Numbers: $\alpha = 1/3$, $\rho = 0.04$, $\delta = 0.06$; ψ is set so that $h = 1/3$ when $\tau = 0.3$.

a) (3 points) Derive the household's optimality conditions (intertemporal and intratemporal), the firm's conditions and the government budget, and write the equilibrium as a dynamic system in (K, c).

Step — household. Budget $\dot{k} = rk + (1-\tau)wh + T - c$. Over a short period Δ the Euler condition is $u_c(t) = e^{-\rho\Delta}(1 + r\Delta)u_c(t+\Delta)$; with $u_c = 1/c$ and $\Delta \to 0$: $\dot{c}/c = r - \rho$. Within each instant, the marginal rate of substitution equals the after-tax wage: $\frac{\psi/(1-h)}{1/c} = (1-\tau)w$.

Step — firms and government. $r^K = \alpha K^{\alpha-1}h^{1-\alpha}$, $w = (1-\alpha)K^\alpha h^{-\alpha}$; no arbitrage: $r = r^K - \delta$; $T = \tau w h$.

Step — system. Goods clearing: $\dot{K} = K^\alpha h^{1-\alpha} - c - \delta K$; Euler: $\dot{c} = c(\alpha K^{\alpha-1}h^{1-\alpha} - \delta - \rho)$; hours are defined implicitly by $\psi c/(1-h) = (1-\tau)(1-\alpha)K^\alpha h^{-\alpha}$, so $h = h(K, c)$ with $h_K > 0$ and $h_c < 0$ (implicit function theorem).

Ref: Kurlat (2020), §9.1 and §9.3; Romer (2012), §2.2.

Grading rubric.

- Euler and intratemporal conditions: 1.5 pts.
- Firms, no-arbitrage, government: 0.5 pt.
- System with $h(K, c)$ and its signs: 1 pt.

b) (3 points) Find the steady state. Which ratios does the tax leave unchanged? Find hours in closed form and calibrate ψ.

Step — ratios. $\dot{c} = 0$: $\alpha(K/h)^{\alpha-1} = \rho + \delta \Rightarrow K/h = (\frac{\alpha}{\rho+\delta})^{\frac{1}{1-\alpha}} = 6.086$, $y/h = 1.826$; $\dot{K} = 0$: $c/h = y/h - \delta K/h = 1.461$, $c/y = 1 - \frac{\delta\alpha}{\rho+\delta} = 0.8$. None depends on τ: r and w are unchanged too.

Step — hours. Intratemporal condition with $c = h(c/h)$ and $w = (1-\alpha)y/h$:

$$\frac{h}{1-h} = \frac{(1-\tau)(1-\alpha)}{\psi\, c/y}.$$

Unique, since the left side rises from 0 to ∞ on $(0, 1)$. With $h = 1/3$ at $\tau = 0.3$: $\psi = \frac{0.7 \cdot 2/3}{0.5 \cdot 0.8} = 7/6$.

Economic intuition. This is Prescott's formula of List B, now with c/y determined in general equilibrium by impatience and depreciation.

Ref: Kurlat (2020), §9.3, p. 177, and Exercise 7.5.

Grading rubric.

- Ratios pinned by $\rho + \delta$: 1.5 pts.
- Hours in closed form and ψ: 1.5 pts.

c) (3 points) The tax rises permanently and unexpectedly from 0.30 to 0.45. Find the long-run changes in h, K, Y, C, r and w.

Step — hours. $\frac{h}{1-h} = \frac{0.55 \cdot 2/3}{(7/6) \cdot 0.8} = 0.393$, so $h = 0.2821$: -15.4%.

Step — scale. K, Y and C are fixed multiples of h: all fall 15.4% (K from 2.029 to 1.717, c from 0.487 to 0.412). K/h is unchanged, so r and w do not move in the long run.

Economic intuition. The tax shrinks the economy without changing its proportions: fewer hours make capital less productive, and households keep only the capital whose return still covers $\rho + \delta$.

Ref: Kurlat (2020), §9.3; Romer (2012), §2.7.

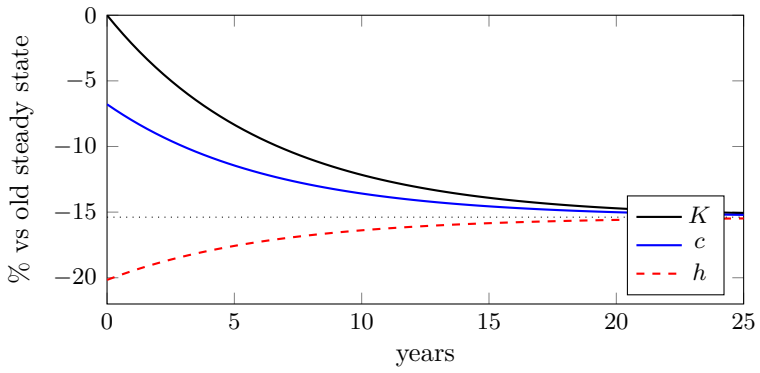
Grading rubric.

- Hours: 1 pt.
- Scale of K, Y, C and unchanged prices: 2 pts.

d) (3 points) Characterize the transition. Linearize the system at the new steady state (use the implicit function for h), show that it is a saddle, find the half-life, the impact responses of c and h, and draw the paths of K, c and h (percent deviations from the old steady state).

Step — Jacobian. Differentiating $\dot{K}$ and $\dot{c}$ with $h = h(K, c)$ (chain rule, $h_K = 0.0754$, $h_c = -0.943$): $J = \begin{pmatrix} 0.1318 & -2.1475 \\ -0.0087 & -0.0918 \end{pmatrix}$. $\det J < 0$: real roots of opposite sign, $\mu_- = -0.1563$, $\mu_+ = 0.1963$; one predetermined (K) and one jump variable (c): a saddle with a unique convergent path. Half-life $\ln 2/0.1563 = 4.4$ years.

Step — impact. Stable arm: $dc/dK = 0.134$. K starts at the old 2.029, so $c(0) \approx 0.412 + 0.134(2.029 - 1.717)$: -6.8% against the old steady state (-7.0% on the exact nonlinear path). Hours jump to -19.8%: they overshoot the long-run -15.4%.



Step — reading. Capital is high relative to the new hours, so consumption stays above its new level; the wealth effect pushes leisure up and hours undershoot. As capital runs down, consumption falls and hours recover to -15.4%.

Ref: Romer (2012), §2.6, pp. 67–71; Kurlat (2020), §9.3.

Grading rubric.

- Jacobian with the implicit h: 1 pt.
- Saddle argument, half-life, impact responses: 1 pt.
- Graph: three paths starting at the right values and converging to -15.4%, h below the line: 1 pt.

e) (3 points) *Parameter analysis.* How do steady-state hours and the speed of convergence depend on τ and ψ? Prove the signs you can, fill the table for $\tau \in \{0, 0.3, 0.45, 0.6\}$, give the limits, and explain why the speed changes.

Step — signs. From (b), $h/(1-h)$ is proportional to $(1-\tau)/\psi$: $\partial h/\partial \tau < 0$, $\partial h/\partial \psi < 0$; K, Y, C move with h; r, w, c/y do not depend on τ or ψ. Limits: $\tau \to 1$ or $\psi \to \infty$ gives $h \to 0$; $\psi \to 0$ gives $h \to 1$.

Step — table.

τ	0	0.30	0.45	0.60
h	0.417	0.333	0.282	0.222
stable root	-0.143	-0.151	-0.156	-0.163
half-life (years)	4.8	4.6	4.4	4.2

Step — why the speed changes. The speed depends on τ and ψ only through steady-state hours ($\psi = 1.5$ at $\tau = 0.3$ gives $h = 0.28$ and the same root as $\tau = 0.45$). Lower h means a higher Frisch elasticity $(1-h)/h$: hours absorb more of the adjustment, so capital reaches its new level faster.

Ref: Kurlat (2020), §7.4 (Frisch elasticity) and §9.3.

Grading rubric.

- Signs and limits proved: 1.5 pts.
- Table: 1 pt.
- Explanation through hours and the Frisch elasticity: 0.5 pt.

Block 2 — Money, inflation and fiscal policy

(Kurlat chs. 10–11; Romer ch. 11 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Money demand × welfare.* Baumol–Tobin: income $Y = 1200$ a year, cost $F = 2$ per trip to the bank, nominal rate $i = 6\%$. The welfare cost of $i > 0$, measured as the Bailey area $\int_0^i m(x)\,dx - i\,m(i)$, is:

A. 8.49

B. 16.97

C. 4.24

D. 0

Answer: A

Step — demand. Minimize $FN + iY/(2N)$: $N^* = \sqrt{iY/(2F)} = 4.243$, $m = \sqrt{FY/(2i)} = 141.4$.

Step — Bailey area. $2\sqrt{FYi/2} - \sqrt{FYi/2} = \sqrt{FYi/2} = 8.49 = FN^*$: the resources spent on trips. The forgone interest $i\,m$ is a transfer to the issuer, not a loss; B adds it.

Ref: Kurlat (2020), §10.4, pp. 199–201; §11.4.

b) (3 points) *Money × expectations.* Cagan with perfect foresight, in logs: $m_t - p_t = -a(p_{t+1} - p_t)$, $a = 4$ (monthly). At $t = 0$ the central bank credibly announces a permanent 10% rise in the money supply at $t = 4$. On impact, p_0 rises by:

A. 0%

B. 4.1%

C. 10%

D. 8.0%

Answer: B

Step — forward solution. $p_t = \frac{m_t}{1+a} + \frac{a}{1+a}p_{t+1}$; with no bubbles $p_t = \frac{1}{1+a}\sum_{j\geq 0}(\frac{a}{1+a})^j m_{t+j}$, so $\Delta p_0 = (\frac{a}{1+a})^4 \cdot 10\% = 4.1\%$.

Ref: Kurlat (2020), Exercise 11.6; Romer (2012), problem 11.1.

c) (2 points, T or F) In the Cagan model with perfect foresight, a credible announcement that money growth will fall permanently in the future leaves inflation unchanged until money growth actually falls.

Answer: F

Step — forward-looking. A lower future money path lowers p_t today: expected inflation falls, real money demand rises and inflation falls on announcement (Question 4(d) runs the same logic in reverse).

Ref: Romer (2012), problem 11.1.

d) (2 points, T or F) *Money × taxes.* If nominal interest income is taxed at 30%, keeping the after-tax real return constant requires the nominal rate to rise by more than one point per point of expected inflation.

Answer: T

Step — Darby effect. $(1-t)i - \pi^e = \text{const} \Rightarrow di/d\pi^e = 1/(1-t) = 1.43$.

Ref: Kurlat (2020), §11.4.

Question 4. (15 points) Discursive. *Money × fiscal policy × expectations: unpleasant arithmetic.* Real money demand (share of GDP) is $m = \kappa e^{-a\pi}$, $\kappa = 0.3$, $a = 2.5$ (annual); output does not grow, so seigniorage in a steady state is $S = \pi m$. Debt (share of GDP) evolves as $b_{t+1} = (1+r)b_t + d - S_t$, with $r = 0.05$, primary deficit $d = 0.005$ and $b_0 = 0.3$. For T years the central bank holds seigniorage at S_1; afterwards debt is frozen and seigniorage covers the interest bill.

a) (3 points) Characterize the steady states of this economy: the relation between debt, deficit and seigniorage, the maximum seigniorage, and the inflation rate if debt is stabilized at once.

Step — budget. Constant debt: $b = (1+r)b + d - S \Rightarrow S = d + rb$.

Step — Laffer. $S(\pi) = \pi\kappa e^{-a\pi}$, $S' = \kappa e^{-a\pi}(1 - a\pi)$: rises up to $\pi = 1/a = 40\%$, $S_{\max} = \kappa/(ae) = 4.41\%$ of GDP, then falls; each $S < S_{\max}$ has two inflation rates. Take the low one (the high one is the wrong side of the Laffer curve).

Step — immediate stabilization. $S_0 = d + rb_0 = 0.02$: $\pi_0 = 8.18\%$.

Ref: Kurlat (2020), §11.3, eq. (11.3.1), p. 216; Romer (2012), §11.9.

Grading rubric.

- $S = d + rb$: 0.5 pt.
- Laffer curve and $S_{\max}$ derived: 1.5 pts.
- π_0 on the low branch: 1 pt.

b) (3 points) Find debt at T and the seigniorage needed afterwards in closed form. Compute them and the inflation rates during and after the tight phase for $S_1 = 0.01$, $T = 10$.

Step — debt. Iterating $b_{t+1} = (1+r)b_t + (d - S_1)$: $b_T = (1+r)^T b_0 + (d - S_1)\frac{(1+r)^T - 1}{r} = 0.4258$.

Step — seigniorage later. $S_2 = d + rb_T = (1+r)^T(d + rb_0 - S_1) + S_1 = S_1 + (1+r)^T(S_0 - S_1) = 2.63\%$: the tightness gap compounds at r.

Step — inflation. Tight phase $\pi_1 = 3.65\%$ (low root of $S = 0.01$); afterwards $\pi_2 = 11.76\%$, against 8.18% with immediate stabilization.

Ref: Kurlat (2020), §11.3.

Grading rubric.

- b_T: 1 pt.
- S_2 in the compounding form: 1 pt.
- Both inflation rates: 1 pt.

c) (3 points) Prove that tighter money today means higher inflation later, and find the length of the tight phase beyond which no steady state exists.

Step — unpleasant arithmetic. $S_2 - S_0 = [(1+r)^T - 1](S_0 - S_1) > 0$ whenever $S_1 < S_0$ and $T \geq 1$. On the low branch π is increasing in S, so $\pi_2 > \pi_0$: lower inflation now is bought with higher inflation forever after.

Step — collapse. No steady state exists once $S_2 > S_{\max}$: $(1+r)^T > \frac{S_{\max} - S_1}{S_0 - S_1}$, so

$$T^* = \frac{\ln[(S_{\max} - S_1)/(S_0 - S_1)]}{\ln(1+r)} = 25.2 \text{ years for } S_1 = 0.01.$$

Beyond T^* the interest bill cannot be paid with seigniorage at any inflation rate.

Ref: Kurlat (2020), §11.3; Romer (2012), §11.9.

Grading rubric.

- Sign of $S_2 - S_0$ and the monotone branch: 1.5 pts.
- T^* derived and computed: 1.5 pts.

d) (3 points) Now money demand is forward-looking, $m_t - p_t = -a(p_{t+1} - p_t)$ in logs, money grows at μ_1 until T and at μ_2 afterwards, and this is known. Find inflation in the years before T and evaluate it 1, 3 and 5 years before the switch with $\mu_1 = 3.65\%$ and $\mu_2 = 11.76\%$.

Step — forward solution. $p_t = \frac{1}{1+a}\sum_{j\geq 0}\lambda^j m_{t+j}$, $\lambda \equiv \frac{a}{1+a}$. Write $m_{t+j} - m_t = \mu_1 j + \Delta \max(0, t+j-T)$ with $\Delta = \mu_2 - \mu_1$. Using $\frac{1}{1+a}\sum_j j\lambda^j = a$: $p_t = m_t + a\mu_1 + a\Delta\lambda^{T-t}$ for $t \leq T$.

Step — inflation. $\pi_s = p_s - p_{s-1} = \mu_1 + a\Delta(\lambda^{T-s} - \lambda^{T-s+1}) = \mu_1 + \Delta\lambda^{T-s+1}$, since $a(1-\lambda) = \lambda$; after T, $\pi = \mu_2$.

Step — numbers. $\lambda = 0.714$: one year before, 9.44%; three years, 6.61%; five years, 5.16%, all above $\mu_1 = 3.65\%$.

Economic intuition. Expected inflation reduces money demand now, so the price level starts rising before the money does: tight money does not even buy low inflation in the meantime.

Ref: Romer (2012), problem 11.1; Kurlat (2020), Exercise 11.6.

Grading rubric.

- Forward solution with the switch: 1.5 pts.
- Inflation path: 1 pt.
- Numbers: 0.5 pt.

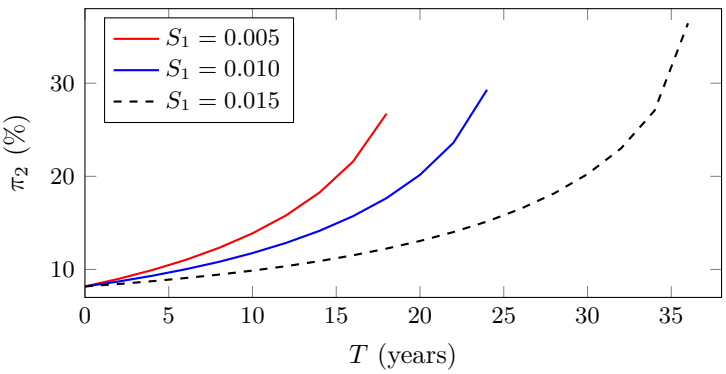
e) (3 points) *Parameter analysis.* How does post-stabilization inflation depend on S_1, T and r? Prove the signs, fill the table for $S_1 \in \{0.005, 0.01, 0.015\}$ and $T \in \{5, 10, 20\}$, compute T^* for each S_1, and draw π_2 against T for the three S_1.

Step — signs. $\partial S_2/\partial T = (1+r)^T \ln(1+r)(S_0 - S_1) > 0$ iff $S_1 < S_0$; $\partial S_2/\partial S_1 = 1 - (1+r)^T < 0$; $\partial S_2/\partial r > 0$ (both $(1+r)^T$ and $S_0 = d + rb_0$ rise). π_2 inherits these signs on the low branch. T^* rises with S_1.

Step — table. π_2 (%):

	$T = 5$	$T = 10$	$T = 20$
$S_1 = 0.005$	10.45	13.88	none
$S_1 = 0.010$	9.66	11.76	20.16
$S_1 = 0.015$	8.90	9.88	13.06

$T^* = 19.7$, 25.2 and 36.1 years.



Step — reading. The tighter the money and the longer it lasts, the higher the inflation that follows, and the curve steepens as S_2 approaches $S_{\max}$, where the low and high equilibria merge. With r

above growth, monetary restraint without fiscal adjustment postpones inflation and enlarges it: the Brazilian 1980s in one equation.

Ref: Kurlat (2020), §11.3; Romer (2012), §11.9.

Grading rubric.

- Signs proved: 1 pt.
- Table and T^*: 1 pt.
- Graph: three increasing convex curves from 8.18%, ordered by S_1, ending at T^*: 1 pt.